

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I K.Manoj Kumar Reddy/192425266 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

manoj

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	

5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

1. Construct three address code for the following.

if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

Step-1:- Condition identification

$(a < b) \text{ and } (c < d) \text{ or } (a > d)$

Condition:

$a < b$

$c > d$

$a > d$

39/40

Step-2:- Table creation

$L_1 \quad L_2 \quad L_3 \quad L_4 \quad L_5 \quad L_6$

Step-3:- TAC for condition

L_1 :

if $a < b$ go to L_2

go to L_5

L_2 :

if $c < d$ go to L_4

go to L_3

L_3 :

if $a > d$ go to L_4

go to L_4

if $a < b$ is false, the complete AND expression is false, so control goes to the else part. If $a < b$ is true expression is evaluated. If either $c < d$ or $a > d$ is true, control goes to the then part.

Step 4: Generate TAC for the THEN part

The statement is $z = x + y * z$. Multiplication has higher precedence

than addition.

$L_4 \rightarrow \text{then } L_7 \rightarrow \text{else}$

THEN:

$Z = x + y + z$

$L_4:$

$t_1 = y * z$

$t_2 = x + t_1$

$Z = t_2$

go to L_6

ELSE:

$Z = Z + 1$

L_5

$t_2 = Z + 1$

$Z = t_2$

$L_6:$

Final TAC:

$L_1:$

if $a < b$ go to L_2

go to L_7

$L_2:$

if $c < d$ go to L_4

go to L_2

$L_2:$

if $a > d$ go to L_4

go to L_7

$L_4:$

$t_1 = y * z$

$t_2 = x + t_1$

$Z = t_2$

go to $L_6:$

2. Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using back patching.

Given boolean expression.

$(a < b)$ and $(c < d)$ or $(e < f)$

Step 1: Apply operator precedence

AND has higher precedence than OR, so the expression is interpreted as:

$[(a < b) \text{ and } (c < d)] \text{ or } (e < f)$

Step 2: Generate incomplete jumps

if $a < b$ go to -

go to -

if $c < d$ go to -

go to -

if $e < f$ go to -

go to -

Step 3: HANDLE AND

$(a < b)$ and $(c < d)$ are true, if $a < b$ is true, control must continue to the second condition

Backpatch statement 1 to statement 3.

1. if $a < b$ goto 3.

Step 4: Handle OR

If the first AND expression is false, the second OR operand $e < f$ must be evaluated

Backpatch statement 2 to statement 5.

Backpatch statement 4 to statement 5.

Step 5: Final Backpatch code

1. if $a < b$ goto 3.

2. goto 5.
3. if $c < d$ go to 7
4. go to 5.
5. if $e < f$ go to 7
6. go to 8
7. TRUE
8. FALSE

3. Construct three address code using the following production for Boolean expression $(a > b)$ or $(c < d)$ or $(e < f)$

$B \rightarrow B_1 \mid B_2$

$B \rightarrow B_1 \mid B_2$

$B \rightarrow (B_1)$

$B \rightarrow \text{true}$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

$[(a > b) \text{ or } (c < d)] \text{ or } (e < f)$

Because this is an or expression, if any one of the three relational condition is true, the composite expression is true.

check the first condition.

L1:

if $a > b$ goto LTRUE

goto L2

check the second condition

L3:

if $c < d$ goto LTRUE

goto L3

check the third condition.

L2:

if $e < f$ goto LTRUE.

goto LFALSE

Final TAC for Q4

L1:

if axb goto LTRUE

goto L2

goto L2

L3:

if c<1 goto LTRUE

goto LFALSE

LTRUE:

goto LEND

LEND.

4. Generate the three address code for the following program fragment

while ($A < C$ and $B > D$) do

if $A = 1$ then $C = C + 1$

else while $A < D$ do

$A = A + B$.

L1:

$t_1 = C + 1$

$C = t_1$

go to L1

L2:

$t_2 = A + B$

$A = t_2$

L2 =

$t_2 = A - B$

if $A == 1$ go to L4

go to L5